

Mahashankh Design & Technology



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Report on The Mahashankh ESP32 Smart Environmental Sentinel

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ABSTRACT:

The Mahashankh ESP32 Smart Environmental Sentinel is a low-power IoT-based environmental monitoring system developed using the ESP32 microcontroller. The system continuously monitors temperature and humidity using a DHT22 sensor and displays the measured values on an SSD1306 OLED display. The ESP32 also transmits the sensor readings through the Serial Monitor for diagnostics and debugging.

To improve energy efficiency, the ESP32 enters Deep Sleep Mode after completing each sensing cycle and wakes up automatically using a timer. The project also utilizes Non-Volatile Storage (NVS) to store sensor values, ensuring that important data persists even after power resets. This design demonstrates the implementation of low-power embedded systems suitable for battery-operated remote monitoring applications.

OBJECTIVE:

The main objectives of this project are:

- 1.Design an ESP32-based environmental monitoring system.
- 2.Read temperature and humidity using the DHT22 sensor.
- 3.Display sensor data on an OLED display.
- 4.Generate alerts using an LED and buzzer when predefined thresholds are exceeded.
- 5.Store sensor values using ESP32 Non-Volatile Storage (NVS).
- 6.Implement Deep Sleep mode to minimize power consumption.
- 7.Demonstrate serial telemetry for monitoring and debugging.

HARDWARE COMPONENTS:

S.NO	COMPONENTS	QUALITY
1.	ESP32 DevKit V1	1
2.	DHT22 Temperature & Humidity Sensor	1
3.	SSD1306 OLED Display	1
4.	LED	1
5.	Piezo Buzzer	1
6.	Breadboard	1
6.	Jumper Wires	1

SOFTWARE COMPONENTS:

1. Arduino IDE
2. Wokwi Simulator
3. GitHub
4. ESP32 Arduino Core
5. Adafruit DHT Library
6. Adafruit SSD1306 Library
7. Adafruit GFX Library

WORKING PRINCIPLE:

The ESP32 initializes the DHT22 sensor, OLED display, LED, buzzer, and the Preferences library used for Non-Volatile Storage. Temperature and humidity values are read from the DHT22 sensor and displayed on the OLED screen while simultaneously being transmitted to the Serial Monitor.

If the temperature exceeds the predefined threshold, the LED and buzzer are activated to alert the user. The latest sensor readings are then stored in NVS to preserve data across resets. After completing the sensing operation, the ESP32 configures a timer wake-up source and enters Deep Sleep Mode to significantly reduce power consumption. When the timer expires, the ESP32 wakes up, restarts the program, and repeats the monitoring cycle.

FEATURES:

Temperature Monitoring

Humidity Monitoring

OLED Display

Serial Telemetry

LED Alert System

Piezo Buzzer Alert

Deep Sleep Power Saving

Non-Volatile Storage (NVS)

Timer-Based Wake-Up

Low Power Embedded Design

CIRCUIT DIAGRAM :

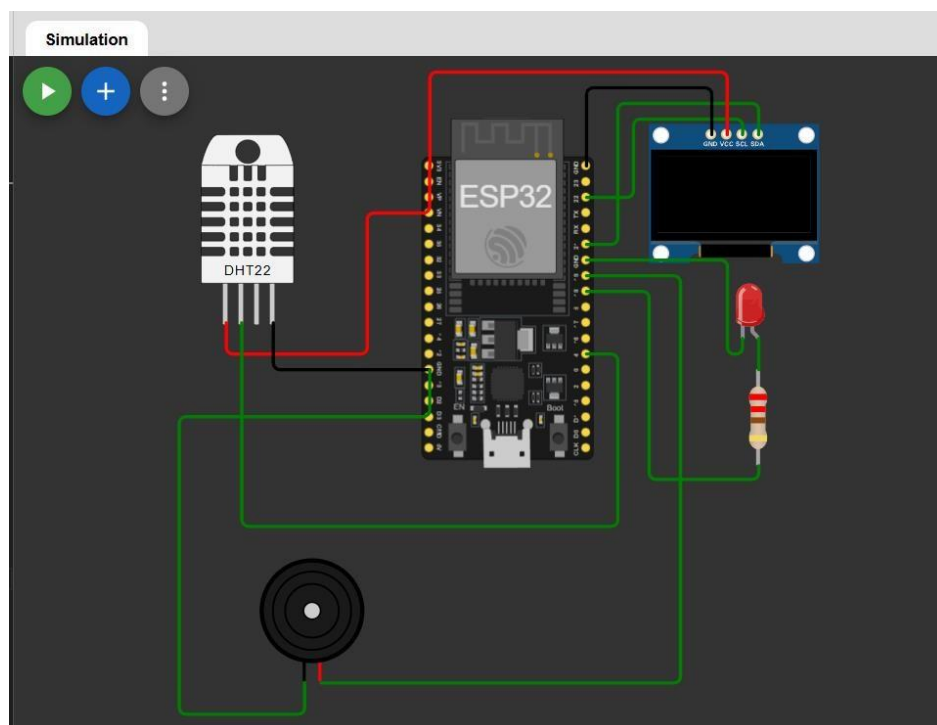


Fig1: Circuit Diagram of Smart Environmental Sentinel

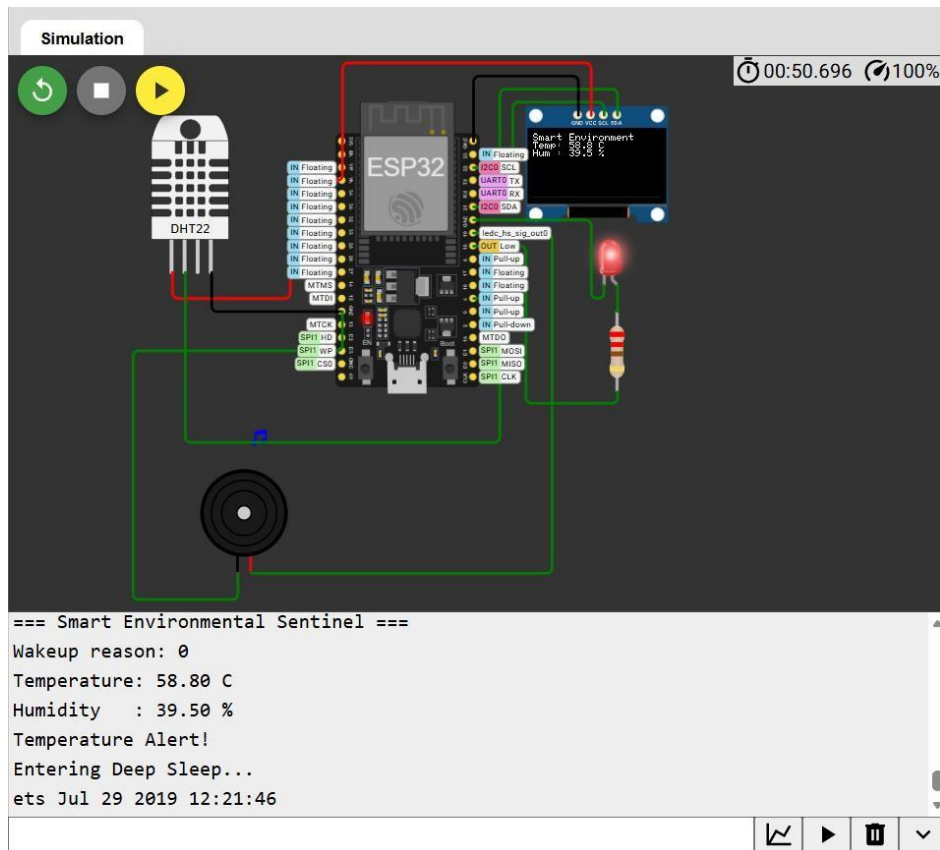


Fig2: Output1 of Smart Environmental Sentinel

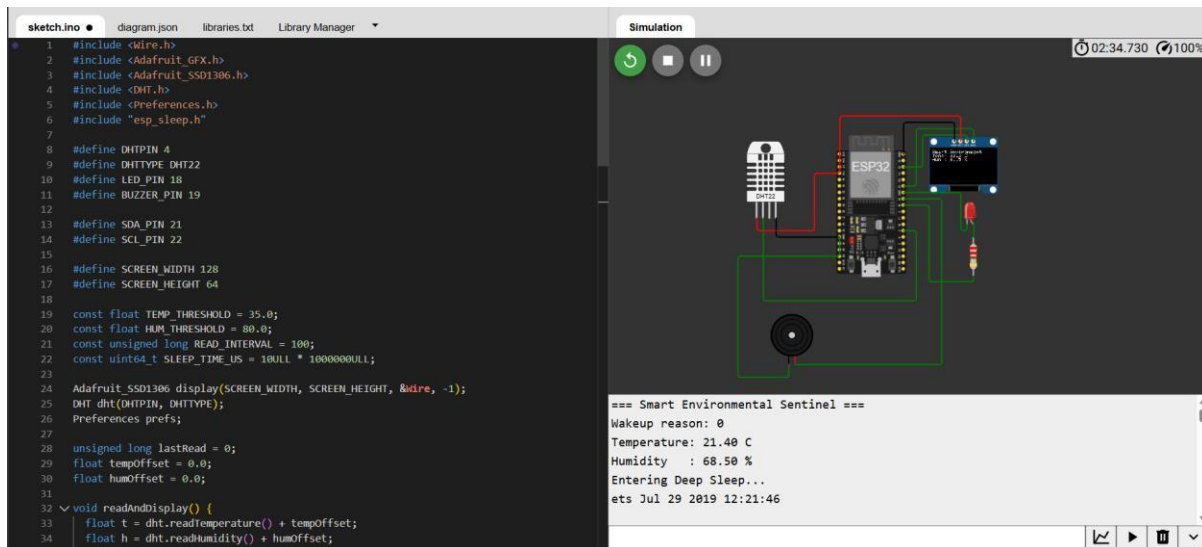


Fig3: Output2 of Smart Environmental Sentinel

APPLICATIONS:

Smart Agriculture

Weather Monitoring

Greenhouse Monitoring

Environmental Data Logging

Smart Home Automation

Industrial Environmental Monitoring

Battery-Powered IoT Devices

ADVANTAGES:

Low Power Consumption

Real-Time Monitoring

Persistent Data Storage

Simple Hardware Design

Easy Integration with IoT Platforms

Reliable Environmental Monitoring

Cost-Effective Solution

FUTURE SCOPE:

Wi-Fi-based Cloud Monitoring

Mobile Application Integration

MQTT Communication

Email/SMS Alert System

Solar-Powered Operation

Additional Sensors (Air Quality, Light, Pressure)

Data Logging to SD Card

CONCLUSION:

The Mahashankh ESP32 Smart Environmental Sentinel successfully demonstrates the development of a low-power embedded IoT monitoring system. The project integrates environmental sensing, OLED visualization, serial diagnostics, non-volatile data storage, and Deep Sleep power management using the ESP32 platform. It provides a practical solution for battery-powered environmental monitoring applications while showcasing key embedded systems concepts such as sensor interfacing, power optimization, and persistent storage.

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